Keith M. Flesher

Engineering Student

OBJECTIVE

I have always been fascinated by the role that engineering plays in shaping our world. From the way that infrastructure shapes our daily lives to the innovations that allow us to push the boundaries of what we can achieve, engineering is at the forefront of progress and is the future.

CERTIFICATIONS

- **CPR Certified**
- **Adobe Illustrator**

SKILLS

- **Leadership**
- **Collaboration**
- **CAD:** Onshape, Fusion 360
- **3D Printing & Prototyping**
- **Mechanical Assembly & Fabrication**
- **Robotics:** FTC, FRC, FLL
- **Technical Teaching & Mentorship**
- **Scouting & Data Collection**
- **Subsystem Prototyping**
- **Project Planning & Coordination**
- **PC Building**

WORK EXPERIENCE

Tri-lakes Irrigation

Fabricated, installed, and repaired sprinkler manifolds; executed trench excavation and maintained irrigation systems; and updated the company logo for modern branding.

Bear Creek Elementary—District 38

Employed to teach robotics to elementary and middle school students; led hands-on instruction using LEGO Robotics kits; introduced core concepts in engineering and coding as part of a paid STEM education program.

Private Commission—Aircraft Design

Commissioned to design and build a 1:20 scale model of a 1984 Luscombe 8A, I conducted research into original blueprints & manuals, developed detailed CAD designs, and fabricated components to closely replicate the aircraft's structure and features.

EDUCATION

Palmer Ridge high School, Monument, Colorado—High School Diploma (Expected May 2026)

Completed all graduation requirements early (May) 2025 to dedicate senior year to advanced STEM and CTE studies.

National Honor Society member

- **Education & Coursework:**
- **Completed:** Honors Biology, Chemistry, Physics, AP Precalculus, PLTW Coding II, PLTW Principles of Engineering & Aerospace Engineering
- **Planned courses: (2025–2026):** AP Physics; AP Calculus; CTE Capstone; Welding

BOY SCOUTS OF AMERICA

Cub Scouts (2014–2019) Boy Scouts (2019–present)

Eagle Scout (2023) at age 14; set a troop record with 47 merit badges in STEM, leadership, and outdoor skills.

Quartermaster, in charge of troop equipment—inventory management, maintenance scheduling, and distribution of gear for camping and events.

Patrol Leader & Senior Patrol Leader, leading weekly meetings for 20+ Scouts and planning trips and community service.

BSA-certified sign language interpreter; CPR & first aid trained; completed **National Youth Leadership Training** (NYLT).

Brotherhood member of the **Order of the Arrow**, mentoring new Scouts in outdoor skills and Scouting values.

Contributed **129+ hours** of service to conservation projects, community cleanups, honor guard ceremonies, food drives, national park maintenance, and church activities.

ROBOTICS ENGINEERING & LEADERSHIP

Mechanical Lead, FTC Team #19970 (2022–2023)

Led the build team in concept design and sketching; designed and constructed drivetrain and shooter mechanisms; served as backup driver and human player, assisting the team in advancing to semifinals at the Denver State Championship and 2nd place Innovate Award.

CAD Instructor & 3D Print Instructor, Monumental Impact Internship

Summer 2023 (142 hrs) & Summer 2024 (60+ hrs): repaired and assembled 3D printers for prototype fabrication; taught robotics fundamentals and CAD (Onshape); supported BattleBots builds and July 4th community STEM events; and engaged with local engineering firms to learn industry design and manufacturing practices.

FLLC Mentor, Teams #62185, #62184, and #62185

Coached middle school students in robotics design, construction, and Python-based code debugging.

Fabrication Lead, FRC Team #4068 (2023–2024)

Coordinated part fabrication, developed CNC operating instructions, helped improve quality, and assisted in the achievement of the Excellence in Engineering Award at the 2024 Utah regional event.

Mechanical Lead, FRC Team #4068 (2025)

Prototyped critical subsystems and oversaw part fabrication. Helped to assemble drivetrain and manipulator assembly. Was a contributor in scouting data collection; 2025 Oklahoma Regional winner. The team placed 82nd worldwide out of 3690, 63rd in the USA out of 2927, and 2nd in Colorado out of 36.

FlesherKm@gmail.com 719-304-4286